

2(a)(i)	(use an) identification key / app / expert ;	1	ignore photographs unqualified A (guide) book / AI / database ignore features – spots / colouring / patterns
2(a)(ii)	any two from: 1 length of (belt) transect ; 2 same / one, person counting (the butterflies on one transect) ; 3 time of the day ; 4 time of the year / season / month ; 5 (period of) time spent counting ;	2	accept any stated length of transect ignore width / size / orientation of belt transect ignore weather ignore time unqualified

Question	Answer	Marks	Guidance
2(a)(iii)	any three from: <i>miscounting:</i> 1 butterflies can fly so difficult to count / move from place to place therefore may counted more than once / could miss some ; 2 butterflies maybe, small / camouflaged, so difficult to see so, inaccurate count /miss some ; 3 difficult to judge 2.5 m distance leading to inaccurate numbers ; 4 person sampling maybe walking too fast so could miss some ; <i>misidentifying:</i> 5 butterflies, can fly / maybe (very) small, therefore unable to see, features / patterns, so inaccurate identification / difficult to identify ; 6 some butterflies have similar, colours / patterns, so could misidentify ; 7 person sampling may, lack the knowledge / AW, so could misidentify ; 8 person sampling may not be able to, see clearly / AW, so could misidentify; 9 person sampling may be walking too fast so could misidentify ;	3	ignore human error AW for miscounting – inaccurate numbers / incorrect numbers / counting or recording, higher / lower, numbers / counting more than once / missing some (BOD – without <i>ref. to</i> counting or numbers) allow inaccurate results for inaccurate numbers AW for misidentification – inaccurate / incorrect identification / hard to identify, wrong identification

Question	Answer				Marks	Guidance
2(b)			prediction		2	ignore predators / pollution ignore light ignore selection pressures 1 ignore greenery
			rural / farmland and woodland	urban / towns and cities		
	1	food / plants / flowers vegetation	more... variety of..	less... less variety of ..		
	2	habitats / niche / trees	more	less		
	3	gardens / parks	less	more		
	4	active conservation of butterflies qualified	less/more	more / less		
	5	pesticides qualified	less/more	less / more		
	6	habitat fragmentation	less	more		
	7	temperature	colder	warmer		
2(c)	max 1 (mp2 only) if correct answer with no working shown 1 correct working: $(13\,500 - 2300) \div 2300 (\times 100)$ or $11\,200 \div 2300 (\times 100) ;$ 2 correct answer / 487 (%) ;				2	no ecf for incorrect working allow alternative method: % calculated –100 ecf incorrect values taken from the graph and all three figures shown correct formula and correct answer from their figures R if not 3 significant figures